# SOFTWARE REQUIREMENTS SPECIFICATION

ISU Infirmary Logbook Management and Health Surveillance System

for the University Health Service, Isabela State University – Echague Campus

Software Engineering 1

| Document Item | Details |
|---|---|
| Intended Organization | University Health Service (UHS), Isabela State University – Echague Campus. |
| System Type | Locally hosted web-based information system on a LAN; FastAPI backend, React frontend, MariaDB/MySQL database. |
| Primary Users | Administrator, Nurse. |
| System Goal | Record patient visits, medicines, and supplies accurately; keep inventory current; give the clinic fast access to patient history and reports; and warn the nurses early when an illness is spreading in a college. |
| Current Delivery State | All modules work end to end (46 API endpoints, 22-table database): patient and visit records, inventory and restock requests, Reports (12 analytics pages, 11 tables, monthly supply report), health surveillance (outbreak early warning), activity logging, maintenance mode, trash/restore, backup/export/restore with an automatic 11:59 PM backup, and password reset by emailed code. Tested with one year of demo data (3,000 patients, 8,901 visits). Clinic-document co-signature (FR-DOC-02) remains implementation work. |
| Commercial State | Non-commercial. Prepared as an academic project for Software Engineering 1. |

## 1. Introduction

| Item | Specification |
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| Purpose | Define the functional, interface, data, security, performance, and quality requirements for the design, implementation, testing, and acceptance of the ISU Infirmary Logbook Management and Health Surveillance System. |
| Problem Context | Patient visits, medicines dispensed, supplies used, and stock levels are recorded by hand. Manual records slow down retrieval of patient history, hide stock shortages until they happen, and make periodic reporting time-consuming. |
| Proposed Solution | A web application hosted on a clinic laptop and used over the LAN. Nurses record patients and visits; medicines and supplies are deducted from inventory automatically; special cases are flagged; restock requests pass from the Nurse to the Administrator for approval; the system produces clinic documents, dashboards, analytics, tables, and reports; a health-surveillance model compares each week's cases per college with the normal level and alerts the nurses on duty to a possible outbreak; and the Administrator has backups, settings, and an activity log. |
| Out of Scope / Future Work | Cloud or internet hosting; network access from the Administration building; SMS notification; email other than password-reset codes; integration with other university systems; patient self-service access. |

### 1.1 Definitions and Acronyms

| Term | Meaning |
|---|---|
| Administrator | Authorized user who manages accounts, system settings, backups, and restock request approval. Has no clinical screens. |
| Nurse | Authorized user who records patients, visits, special cases, inventory, and clinic documents. |

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| UHS | University Health Service; the office that runs the campus infirmary. |
| NASA | Non-Academic Staff; non-teaching university personnel. |
| PWD | Person with Disability. |
| Special Case | Patient flag that needs priority or careful handling (e.g., PWD, Allergy, Asthma, Hypertension). |
| Disposition | Outcome recorded when a visit is signed out (e.g., Back to Class/Work, Sent Home, Referred to Doctor). |
| Ledger | Audit record of every stock movement (inventory_transactions). |
| Soft Delete | Marking a record as deleted so it stays recoverable, instead of removing it. |
| Activity Log | Record of who did what and when (account, record, inventory, security, and backup events). |
| Maintenance Mode | Setting that blocks Nurse sign-in, with a notice, while the Administrator does maintenance. |
| Reset Code | Six-digit, single-use code emailed to a user to confirm a password reset. |
| Health Surveillance | Weekly monitoring of watched illnesses (Fever, Viral Flu, Cough / Colds, Stomach Pain, Allergic Reaction) per college to detect a possible outbreak early. |
| Outbreak Alert | Notice on the Nurse Dashboard, e.g. "Possible outbreak: Viral Flu in CCSICT – 18 cases this week", raised when a week's count is far above the expected (normal) level. |
| Poisson Model | Statistical model (Poisson regression) that gives the expected weekly cases for a college, illness, and time of year; stored in health_surveillance/model.json. |
| App Password | 16-character Gmail password that lets the server send email without the account's normal password. |
| JWT | JSON Web Token; signed token that proves a user is signed in. |
| LAN | Local Area Network. |
| SMTP | Simple Mail Transfer Protocol; used to send reset codes through Gmail. |

## 2. Scope and System Functions

| Module / Function | Scope / Description | Primary User(s) | Main Output |
|---|---|---|---|
| Authentication & Role Access | Username/password sign-in with role-based access, account lockout after repeated failed attempts, and password reset by a code sent to the user's email. | Admin, Nurse | Authorized session and role-appropriate screens |
| Dashboards & Reports | Nurse: outbreak and low-stock alerts, today's visits, referrals, visit trend, top complaints, top medicines, busiest hours, client breakdown, visit outcomes, active patient queue. Admin: backup status, accounts, stock needing attention, pending requisitions, sign-in activity, account activity, campus complaint summary, stock status. | Admin, Nurse | Monitoring summaries, charts, and CSV reports |
| Patient Records & Special Cases | Register, search, edit, and soft-delete patients; flag and retire special cases. | Nurse | Patient profile, flags, and visit history |
| Visit Records | Record complaints, vital signs, management, medicines, and supplies; sign out with a disposition. | Nurse | Complete visit record with stock deducted |
| Search & History | **Search patients by name, ID number, department, program, year level, flag, complaint, medicine given, disposition, or attending nurse; filter by patient type, department, flag, and date.** | Nurse | Filtered visit history |
| Stock & Supplies | Maintain medicine and supply inventory with automatic stock status and ledger. | Nurse | Inventory monitor |

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| Restock Requests | Nurse requests stock; Administrator approves or denies; delivery is received into stock. | Nurse, Admin | Tracked restock requests |
| User Management & Settings | Manage accounts, passwords, and security/retention settings; view the Activity Log; switch Maintenance Mode; restore records from Trash. | Admin | Controlled access, configuration, and audit trail |
| Reports & Clinic Documents | Reports group: 12 Analytics pages (one chart each), 11 Tables pages with CSV export, and the Monthly supply report; printable Medical Certificate and Parental Notification. | Nurse, Admin | Charts, tables, CSV files, and printable documents |
| Health Surveillance | Counts this week's visits per college and watched illness, compares them with the model's expected level, and raises outbreak alerts; Outbreak watch chart of the last 8 weeks. | Nurse | Outbreak alert and weekly trend |
| Backup & Recovery | Manual and automatic daily backup, SQL export, backup download, and restore. | Admin | Recoverable backup set |

## 3. User Roles and Privileges

| Role | Allowed Functions | Restriction / Notes |
|---|---|---|
| Administrator | Admin Dashboard, Stock Requests (approve/deny), User Management, Reports, Audit (Activity) Log, System Settings, Maintenance Mode, Trash, backup and recovery functions. | Created by the setup script (first account) or by an existing Administrator. Has no clinical screens and cannot change own role or status. Can still sign in during Maintenance Mode. |
| Nurse | Nurse Dashboard with outbreak alerts, Add Visit Record, Patient Records, Priority & Special Cases, Search & History, Stock & Supplies (including restock requisitions and delivery confirmation), Reports, clinic documents. | Account is created by the Administrator. Cannot manage accounts, change settings, back up or restore, or approve/deny restock requests. Cannot sign in during Maintenance Mode. |

## 4. Functional Requirements

| Requirement ID | Requirement |
|---|---|
| FR-AUTH-01 | Require username and password before any protected function is used. |
| FR-AUTH-02 | Issue a signed access token (JWT) that expires 480 minutes after sign-in; keep the session through a page refresh and end it when the browser is closed. |
| FR-AUTH-03 | Show only the navigation of the signed-in role (Administrator or Nurse). |
| FR-AUTH-04 | Deny nurse-only functions to Administrators and admin-only functions to Nurses at the server. |
| FR-AUTH-05 | Lock sign-in for 15 minutes after 5 failed attempts within 15 minutes; both values are configurable. |
| FR-AUTH-06 | Record each sign-in attempt with username, IP address, time, and result; show the same message for unknown usernames and wrong passwords. |
| FR-AUTH-07 | **Provide password recovery by Gmail code from "Forgot password?" in three steps: (1) the user enters the email on the account and the system emails a one-time 6-digit code; (2) the user enters the code (valid 5 minutes, single use, cancelled after 5 wrong tries; an expired code says so and does not count as a try; a new code can be requested after 60 seconds, at most 3 times per 15 minutes); (3) only after the code is accepted, the user sets a new password that follows the password rule. Accounts with no email are reset by the Administrator in User Management.** |
| FR-AUTH-08 | While Maintenance Mode is on, refuse Nurse sign-in and show the Administrator's maintenance notice; Administrators can still sign in. |
| FR-DASH-01 | Display the Nurse Dashboard: outbreak alert banner, low/out-of-stock alert, today's visits, range |

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| | total, referrals today, low/out-of-stock count, Visit Trend (7 days, 30 days, semester), Top Complaints, Active Patient Queue, Top 5 Most Requested Medicines, Busiest Hours, Client Breakdown, and Visit Outcomes, filtered by Today, This Week, This Month, or date range. |
| FR-DASH-02 | **Display the Admin Dashboard: automated-backup status, active accounts, stock needing attention, pending requisitions, a 7-day Sign-in Activity chart (successful sign-ins per day, with failed attempts), Account Activity, Campus Complaint Summary, and Stock Status.** |
| FR-DASH-03 | Show a notification count: low or out-of-stock items for the Nurse; pending restock requests for the Administrator. |

## 4.1 Patient, Special Case, and Visit Records

| Requirement ID | Requirement |
|---|---|
| FR-PAT-01 | Register a patient with ID number, full name, sex, date of birth, civil status, contact number, patient type (Student, Faculty, NASA, Visitor), department, program, and year level. |
| FR-PAT-02 | Reject an ID number already registered to another patient. |
| FR-PAT-03 | **Search patient records by name, ID number, or chief complaint; filter by gender, year level, program, department, and date range; sort by newest, oldest, or name; and return paged results (25, 50, or 100 rows).** |
| FR-PAT-04 | Allow the Nurse to edit a patient and move a patient to trash (soft delete). |
| FR-PAT-05 | Show a Patient Profile with computed age, visit count, last visit, active flags, and visit history; export and print lists. |
| FR-FLAG-01 | Flag a patient as PWD, Senior Citizen, Allergy, Anxiety, Asthma, Hypertension, or Other Special Case with notes; reject a duplicate active flag of the same type. |
| FR-FLAG-02 | Record who flagged the patient and when; allow a flag to be retired while keeping its history. |
| FR-VIS-01 | **Record a visit with patient lookup, chief complaint, blood pressure, temperature, pulse rate, management, and treatment notes; complaint tags (one primary) can be added when editing the visit.** |
| FR-VIS-02 | **Record medicines dispensed (quantity and dosage or instructions) and supplies used (quantity and remarks); show the patient's active flags and on-hand stock during triage.** |
| FR-VIS-03 | Save the visit and its stock deductions as one transaction; when any item is short, save nothing and show the shortage. |
| FR-VIS-04 | **Sign out a visit with a disposition and end time; a new visit starts as In Care and becomes Completed when signed out.** |
| FR-VIS-05 | Allow the Nurse to edit a visit (recording who updated it) and move a visit to trash. |
| FR-SRCH-01 | **Search patients from one search box by name, ID number, department, program, year level, special-case flag, chief complaint, medicine given, disposition, or attending nurse; filter by patient type, department, priority flag, and date range; sort by last visit, name, or most visits.** |

## 4.2 Inventory and Restock

| Requirement ID | Requirement |
|---|---|
| FR-STK-01 | Maintain medicines and supplies with name, category, unit, reorder level, and opening quantity; reject duplicate names. |
| FR-STK-02 | Set stock status automatically: Out of Stock at quantity ≤ 0, Low at quantity ≤ reorder level, otherwise High; filter inventory by status and warn of items near expiry. |
| FR-STK-03 | **Allow manual restock by entering the quantity received (above 0), with an optional new reorder level and expiry date.** |
| FR-STK-04 | Record every stock movement in the ledger with type (released, restocked, adjustment, expired, damaged) and reference (visit, stock request, manual). |
| FR-REQ-01 | Allow the Nurse to submit a restock request with one or more items, quantities, and reasons. |

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| FR-REQ-02 | Allow the Administrator to approve (with approved quantities) or deny (with reason) a pending request. |
| FR-REQ-03 | Allow the Nurse to confirm delivery of an approved request, which adds approved quantities to stock, writes ledger entries, and completes the request in one transaction; reject out-of-sequence status changes. |

### 4.3 Reports, Health Surveillance, Documents, and Administration

| Requirement ID | Requirement |
|---|---|
| FR-RPT-01 | Provide Reports → Analytics with 12 single-chart pages (Visits per month, Visits by day of the week, Busiest hours, Visit outcomes, Frequent complaints, Top requested medicines, Common treatment pairings, Client breakdown, Highest visiting departments, Student demographics, Faculty & staff by age, Outbreak watch), each with a date range (default: semester), gender, year level, program, and department filters, and Print report. |
| FR-RPT-02 | Provide Reports → Tables with 11 table pages (Visits per month, Visit outcomes, Complaints / illnesses, Medicines dispensed, Departments, Common treatment pairings, Client breakdown, Students by year level, Students by gender, Faculty & staff by age, Outbreak watch), each exportable as CSV. |
| FR-RPT-03 | Provide the Monthly supply report: beginning stock, received, issued, and ending stock of every medicine and supply for a selected month, with CSV export. |
| FR-RPT-04 | Display Common Treatment Pairings, reporting a pairing only when it appears in at least 3 visits with a match rate of at least 50%. |
| FR-SURV-01 | Each week (Monday to Sunday, Manila time), count visits per college (CCSICT, CON, COE, CED, CA, CBM, CCJE, CAS, Other) and watched illness (Fever, Viral Flu, Cough / Colds, Stomach Pain, Allergic Reaction), using the visit's primary complaint tag or, when untagged, keywords in the chief complaint (English or Filipino). |
| FR-SURV-02 | Compare each count with the Poisson model's expected level for that college, illness, and time of year, calibrated to the clinic's own last 12 weeks; raise an alert when the count has at least 4 cases and is unlikely under the model ($p < 0.0025$ for this week alone, or for the last 2 weeks together while still elevated). The current week is compared with the part of the week that has passed. |
| FR-SURV-03 | Show each alert to the nurses on the Nurse Dashboard as "Possible outbreak: <illness> in <college>" with the number of cases, without stating the expected level. |
| FR-SURV-04 | Provide Outbreak watch (chart and table): weekly cases of each watched illness for the last 8 weeks next to the normal level, marking the weeks that raised an alert, plus this week's cases per college. |
| FR-SURV-05 | Train and evaluate the model offline from weekly counts (health_surveillance/train.py and evaluate.py); the server only reads model.json and needs no machine-learning libraries. |
| FR-DOC-01 | Generate a printable Medical Certificate (ISUE-UHS-MCR-007) and Parental Notification (ISUE-UHS-PaN-006) on ISU letterhead from a patient's visit record. |
| FR-DOC-02 | Route clinic documents for UHS Director co-signature before release. (Not yet built.) |
| FR-USER-01 | Allow the Administrator to create accounts with an optional email, change role or status, and reset passwords; reject duplicate usernames or emails. |
| FR-USER-02 | Require passwords of 8 to 72 characters with at least one capital letter and one number, plus a symbol when the symbol rule is enabled; show the rules as a checklist while typing; store only bcrypt hashes. |
| FR-SET-01 | Allow the Administrator to set soft-delete retention, failed-login limit, lockout duration, password symbol rule, automated backup on/off, backup retention (days), and Maintenance Mode with its notice. |
| FR-SET-02 | **Record account, record, inventory, security, and backup events in an Activity Log with user, action, and time; let the Administrator search it, filter by type, page through it, and export it as CSV.** |
| FR-SET-03 | Provide a Trash view listing soft-deleted patients and visits with the date their retention period ends, and allow the Administrator to restore them. |

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| FR-BACK-01 | Allow the Administrator to back up now, export the database as a .SQL file, and download any saved backup. |
| FR-BACK-02 | Take one automatic backup every day at 11:59 PM (Manila time) while automatic backup is on, or as soon as the server starts if that time was missed; delete backups older than the backup retention setting (default 30 days). |
| FR-BACK-03 | **Allow the Administrator to restore the database from a selected backup after confirming a warning; take a safety backup automatically before restoring.** |

# 5. Data Requirements

| Data Entity | Minimum Required Information | Purpose |
|---|---|---|
| User | Name, username, email, password hash, role, status, last login, timestamps. | Identity, controlled access, and password-reset email. |
| Login Attempt | Username, linked user, IP address, result, time. | Lockout, sign-in audit, and sign-in chart. |
| Patient | ID number, full name, sex, date of birth, civil status, contact number, patient type, department, program, year level. | Patient identity and demographics. |
| Special Case Flag | Patient, case type, notes, originating visit, flagged by, flagged at, active status. | Priority and careful handling. |
| Visit | Patient, attending nurse, chief complaint, vital signs, management, treatment notes, disposition, start/end time, status. | Clinic encounter record. |
| Visit Complaint / Medicine / Supply | Tagged complaints (primary flag); medicine and quantity, dosage, instructions; supply and quantity; given by/at. | Treatment details and stock usage. |
| Medicine / Supply / Stock | Name, category, unit; quantity, reorder level, expiry date, stock status. | Inventory catalog and levels. |
| Inventory Transaction | Item, transaction type, quantity, reference type and ID, user, time. | Stock audit ledger. |
| Stock Request | Requester, items, requested/approved quantities, reason, status, reviewer, receiver, timestamps. | Restock approval pipeline. |
| Activity Log | User, action, log type (account, record, inventory, security, backup), affected record, time. | Audit trail for the Administrator. |
| Lookup / Setting | Patient types, complaints, dispositions, case types, categories; setting name/value/type. | Reference data and configuration. |
| Backup File | File name with date and time, kind (manual, automatic, safety), size. | Recovery point stored as .sql in the backend backups folder. |
| Reset Code | Hashed 6-digit code, expiry time, wrong-try count; kept in server memory only, never in the database. | Confirms a password reset. |

## 5.1 Data Integrity

| ID | Data Requirement |
|---|---|
| DR-01 | Maintain foreign-key relationships among users, patients, visits, flags, medicines, supplies, stock, ledger, and restock requests (22 tables). |
| DR-02 | Keep patient ID number, username, and email unique; allow one active flag per patient per case type. |
| DR-03 | **Soft-delete patients and visits (moved to Trash) and keep them recoverable for the retention period (default 90 days); user accounts are deactivated instead of deleted.** |
| DR-04 | Store timestamps in UTC and display them in Asia/Manila time; store quantities as DECIMAL. |

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| DR-05 | Compute stock status in the database; change quantities only through the stock service, which writes a ledger entry for each change. |
| DR-06 | Validate required fields and ranges on the server before saving and return readable error messages. |
| DR-07 | Keep backup files out of version control; a restore replaces the whole database with the selected backup. |

## 6. External Interface Requirements

| Interface ID | Requirement |
|---|---|
| UI-01 | Provide browser screens for sign-in, password reset, role dashboards, and all Nurse and Administrator functions, following the final wireframe (figma ver2) and the final mockup (ISU-Infirmary-Frontend-Preview-4.html). |
| UI-02 | Provide readable paged tables, filters, charts, status labels, and modal forms; confirm before moving records to trash or restoring a backup. |
| UI-03 | Show the server's error message after a failed action and a result after each save. |
| API-01 | Frontend communicates with the FastAPI REST API under /api using JSON (camelCase); lists return { items, total, page, perPage }. |
| API-02 | Send the access token as Authorization: Bearer on protected requests; publish API documentation at /docs. |
| API-03 | Serve backup and .SQL export files as authenticated downloads for the Administrator. |
| PRINT-01 | Print Medical Certificate, Parental Notification, Insights report, and record lists through the browser; export lists, reports, and the Activity Log as CSV. |
| EMAIL-01 | Send password-reset codes through Gmail SMTP (smtp.gmail.com, port 587, STARTTLS) using an App Password kept in the backend .env file. |
| NET-01 | Client devices open the server's LAN address on port 5173; the frontend forwards /api to the backend (8000) on the same machine; ports 8000 and 3306 stay closed to the network. The server needs internet only to send reset emails. |

## 7. Non-Functional Requirements

| Category | ID | Requirement / Target |
|---|---|---|
| Performance | NFR-PERF-01 | **Common screens should load within 2 seconds on the LAN with 2 client devices (Reports: about 2 seconds the first time for a semester of visits, then instant); lists are paged (up to 1,000 visits per request for reports; 500 for the inventory list and Activity Log).** |
| Reliability | NFR-REL-01 | Visit saving and restock receiving must complete fully or not at all. |
| Reliability | NFR-REL-02 | Stock updates lock the stock row so simultaneous changes do not overwrite each other; saved records remain after restart. |
| Reliability | NFR-REL-03 | A safety backup is taken before every restore, so a restore can be undone. |
| Security | NFR-SEC-01 | Use bcrypt password hashing, signed JWTs, server-side role checks, and configurable account lockout. |
| Security | NFR-SEC-02 | Keep secret key, database credentials, and the Gmail App Password in an .env file outside version control; use ORM parameterized queries. |
| Security | NFR-SEC-03 | Protect patient health information through role-based access and by exposing only port 5173 on the LAN. |
| Security | NFR-SEC-04 | Reset codes are random, stored only as hashes, valid for 10 minutes, single use, and cancelled after 5 wrong tries; a new code can be requested once every 60 seconds; the reply is the same whether or not the username exists. |

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| Usability | NFR-USA-01 | A trained nurse should complete a routine visit (lookup, complaint, medicine, sign-out) in 3 minutes or less. |
| Compatibility | NFR-COM-01 | Operate on current Chrome, Edge, and Firefox; client devices need no installation. |
| Maintainability | NFR-MNT-01 | **Keep business rules in the services layer, follow consistent naming conventions, and apply schema changes through the versioned schema_v2.sql script.** |

## 8. Hardware and Software Environment

| Component | Requirement / Status | Purpose / Note |
|---|---|---|
| Server Laptop | Windows 10/11 64-bit; Core i5/Ryzen 5; 8 GB RAM (16 GB recommended); SSD; Wi-Fi. | Hosts database, backend, and frontend; also the Administrator workstation. |
| Nurse Device | Laptop or desktop with a modern browser and Wi-Fi. | Nurse workstation in the infirmary. |
| Network | Wireless router or pocket Wi-Fi; DHCP reservation (fixed IP) for the server; internet access for sending reset emails. | LAN shared by server and client devices. |
| Printer / UPS / Backup Drive | Any supported printer; 650–1000 VA UPS; 1 TB external drive. | Documents and reports; power protection; off-laptop copies of backups. |
| Database | MariaDB 10.4+ via XAMPP (MySQL 8 compatible); mysqldump and mysql tools from XAMPP. | Persistent storage of all records; backup and restore. |
| Backend | Python 3.13, FastAPI, SQLAlchemy 2.0, Alembic, Pydantic, python-jose (JWT), passlib/bcrypt; NumPy only for retraining the surveillance model. | Application API, business rules, and migrations. |
| Frontend | Node.js 20.19+ or 22.12+, React 19, Vite 8, ESLint. | Browser interface and API forwarding. |
| Email Account | Gmail account with 2-Step Verification and an App Password. | Sends password-reset codes. |

## 9. Business Rules

| ID | Business Rule |
|---|---|
| BR-01 | Only the Administrator creates accounts; the first Administrator is created with the setup script. |
| BR-02 | The Administrator has no clinical screens, and the Nurse has no administrative screens; an Administrator cannot change own role or status. |
| BR-03 | A visit is saved completely or not at all; insufficient stock for any item blocks the save. |
| BR-04 | Stock quantities change only through the stock service, and every change is written to the ledger. |
| BR-05 | Restock requests follow Pending → Approved or Denied → Completed; only the Administrator approves or denies. |
| BR-06 | A patient cannot hold two active flags of the same type; a Visitor has no ISU ID number. |
| BR-07 | A visit is closed only by signing it out with a disposition; deleted records stay recoverable for the retention period. |
| BR-08 | Only the Administrator backs up, exports, restores, views the Activity Log, restores from Trash, or switches Maintenance Mode. |
| BR-09 | Password reset by email works only for accounts with an email on file; other accounts are reset by the Administrator. |

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| BR-10 | Outbreak alerts go to the nurses on duty, not the Administrator, and are a warning for clinic follow-up, not a diagnosis. |

## 10. Implementation Roadmap Alignment

| Phase | Major Activity | Expected Deliverable | Status (Nov 10) |
|---|---|---|---|
| 1 (Sep 7–11) | Problem definition, interview with the UHS nurse, repository setup | Project concept, interview notes, GitHub repository | Done |
| 2 (Sep 14–18) | Database design, Figma wireframes, backend core, React screens | 22-table schema, Figma design, initial backend and frontend | Done |
| 3 (Sep 21–25) | Backend–frontend integration, LAN setup, SRS | Integrated build (31 endpoints), setup guide, SRS | Done |
| 4 (Sep 28–Oct 2) | Clinic document co-signature, activity logging, redesigned frontend (ver2) | FR-DOC-02, Activity Log (FR-SET-02) | Activity Log done; FR-DOC-02 pending |
| 5 (Oct 5–9) | Backup, export, restore, trash, maintenance mode, reports, password reset by email | FR-BACK-01 to 03, FR-SET-03, FR-AUTH-07, FR-AUTH-08, FR-DASH-02 | Done (43 endpoints) |
| 6 (Oct 12–16) | Testing; health surveillance model; finish FR-DOC-02 | Test results; outbreak early warning (FR-SURV-01 to 05) | Testing and FR-SURV done; FR-DOC-02 pending |
| 7 (Oct 19–23) | User Acceptance Testing with UHS | UAT results and revision list | To be confirmed with UHS |
| 8 (Oct 26–31) | Final revisions, documentation, new frontend (Reports group) | Final build, UML, user manual, final presentation | Done |
| 9 (Nov 2–10) | Backend for the final frontend, 1-year demo data, full system test, GitHub upload | Final system on GitHub with SRS, UML, wireframe, mockup, and evidence screenshots | Done |

## 11. Verification and Acceptance Criteria

| ID | Acceptance Scenario | Expected Result |
|---|---|---|
| AC-01 | Nurse or Administrator signs in with valid credentials. | The role's dashboard and navigation are displayed. |
| AC-02 | Wrong password is entered 5 times within 15 minutes. | Further attempts are refused for 15 minutes. |
| AC-03 | A user calls a function of the other role. | The server refuses the request. |
| AC-04 | Nurse registers a patient with an existing ID number. | Save is refused with a duplicate ID message. |
| AC-05 | Nurse saves a visit with medicines and supplies in stock. | Visit is saved; stock decreases; ledger entries are created. |
| AC-06 | Nurse saves a visit with more medicine than is in stock. | Nothing is saved; stock is unchanged; the shortage is shown. |
| AC-07 | Stock falls to or below the reorder level. | Item shows Low and appears in the Nurse's notifications. |
| AC-08 | Restock request is submitted, approved, and received. | Stock increases by the approved quantities; request is Completed. |
| AC-09 | Nurse prints a Medical Certificate for a visit. | Certificate shows the correct patient details and findings. |
| AC-10 | Administrator takes a backup, changes data, then restores the backup. | Backup file is created; a safety backup is taken; restored records match the backup. |

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| AC-11 | User with an email on file uses Forgot password and enters the emailed code. | New-password fields appear only after the code is accepted; the user signs in with the new password; the code cannot be reused. |
| AC-12 | Wrong reset code is entered 5 times. | The code is cancelled and the user must request a new one. |
| AC-13 | Administrator turns on Maintenance Mode. | Nurse sign-in is refused with the notice; Administrator can still sign in. |
| AC-14 | Nurse moves a patient to trash; Administrator restores it from Trash. | Patient reappears in Patient Records; both actions appear in the Activity Log. |
| AC-15 | A college has many more cases of a watched illness this week than normal (demo data: Viral Flu in CCSICT). | The Nurse Dashboard shows "Possible outbreak: Viral Flu in CCSICT" with the case count; Outbreak watch marks this week. |

## 12. Evaluation of Source Content and Clarifications Needed

| Project Observation | Evaluation | SRS Treatment / Required Action |
|---|---|---|
| Activity logging, trash, maintenance mode, backup/export/restore, Reports, and health surveillance are built; clinic-document co-signature is not. | Implementation matches the requirements except FR-DOC-02. | Build FR-DOC-02 in Phase 6, or confirm the signing process with UHS first (Section 13, item 1). |
| System Settings buttons (Back Up Now, Restore, Export .SQL, Trash) are now connected to the backend. | Resolved. | Verify with AC-10 and AC-14. |
| Settings rows min_password_length, password_reset_email, and backup_schedule exist in the database but are not read by the code (the 8-character minimum is fixed; reset email is turned on by the .env file; the automatic backup runs at 11:59 PM). | Schema is ahead of the code. | Implement or remove these rows before acceptance. |
| Soft-deleted records show the date their retention period ends, but no job removes them after that date. | Retention is shown, not enforced. | Add an automatic purge or keep records until removed manually; confirm with UHS (Section 13, item 3). |
| Password reset sends email from a team member's Gmail account. | Works for development; depends on a personal account. | Switch to a clinic-owned Gmail account before deployment. |
| The infirmary and the Administration building are separate buildings; deployment is LAN only. | The Administrator cannot reach the system from the Administration building. | Keep inter-building access out of scope; confirm with UHS where the Administrator works. |
| Treatment pairings in Insights are statistical associations. | Descriptive only; not a medical diagnosis. | Label the Insights output as descriptive information for clinic planning. |
| **Add Visit Record has no complaint-tag picker (tags can only be set when editing a visit).** | **Surveillance and Reports fall back to keywords in the chief complaint.** | **Add the tag picker to Add Visit.** |
| **Priority & Special Cases now shows who flagged each patient; editing an active flag is refused as a duplicate, and restoring a flag creates a new row.** | **Display fixed; edit behaviour remains.** | **Update flags in place.** |
| **The Administrator's response is shown on the Nurse's requisition view; the form still says pending** | **Screen text promises more than the system does.** | **Add edit/cancel or remove the sentence.** |

| | | |
|---|---|---|
| **requests can be edited or cancelled, but no such function exists.** | | |
| **Faculty & staff by age now shows data (fixed Nov 8). The Visitor ID "assigned automatically" note and the "Sent Home prints a parental notification automatically" note are not implemented; the stock ledger has no screen.** | **Interface is ahead of the code.** | **Implement each item or remove the label before UAT.** |
| **The login footer always says "5 failed attempts / 15 minutes", and the Add User form always requires a symbol, even when System Settings are changed.** | **Screen text does not follow the settings.** | **Read both values from the settings.** |
| **System test with one year of demo data (Nov 8–10): a defect stopped the Administrator from approving restock requests (the page sent item ids in a different format).** | **Fixed and verified (approved quantities, including edited ones, are saved).** | **None; keep in the regression test.** |

## 13. Requirements Requiring Final Confirmation

| No. | Item to Confirm | Why it Matters |
|---|---|---|
| 1 | Who signs and co-signs the Medical Certificate and Parental Notification, and the final form layouts. | Determines the document workflow and printed output. |
| 2 | Where off-laptop copies of backups are kept, and how often they are copied. | Backups on the same laptop are lost if the laptop fails. |
| 3 | Retention period for soft-deleted records under UHS data privacy policy, and whether they are purged automatically. | Protects patient data and sets the purge rule. |
| 4 | Required report types and reporting schedule for UHS. | Ensures reports match actual office needs. |
| 5 | Which email account sends reset codes, and whether every staff account will have an email. | Accounts without email can only be reset by the Administrator. |

## 14. Software Engineering Test Plan Mapping

| Testing Level | What Will Be Verified | Related Areas |
|---|---|---|
| Unit Testing | Individual functions and validation rules work in isolation. | Password rules, lockout counting, stock status rule, clinic date-range conversion, treatment-pairing thresholds, reset-code expiry and try limit, input validation. |
| Integration Testing | Modules exchange data correctly and keep integrity. | Sign-in → token → role check; visit save → stock deduction → ledger; restock request → approval → receipt → stock; forgot password → code → new password → sign-in; backup → restore; action → Activity Log entry. |
| System Testing | Complete system works under realistic scenarios. | Nurse and Administrator workflows over the LAN on two devices, printing, CSV export, maintenance mode, performance, and closed ports. |
| User Acceptance Testing | Intended users can complete their tasks and accept the outcome. | UHS nurse and administrator tasks; usability, reliability, security, and readiness for clinic use. |

### Developers

BSCS 3-1
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